Question Answering with Retrieval Augmented Generation

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Abstract

In this work I build a RAG-assisted QA system using BM25 retriever. Wikipedia pages were collected for evaluation. Using RAG in some cases increased the question answering accuracy from 36% to 100%.

1 Dataset

81 Wikipedia pages were collected and curated question sets were made from their content.

1.1 Page Collection

Pages were scraped using `scrapling` tool and cleaned further using regular expressions, only keeping the main text, dropping links and footer sections.

The pages were from selected themes [1] listed in Table 1. To collect this list, ChatGPT 5.5 was prompted for page suggestion in each category.

The document collection includes both closely related and unrelated pages to test retrieval quality and distract the retriever. The pages include:

- Iranian languages
- Baklava
- Tehran
- Sharks!

Theme	**#**
Language	14
Language family	8
Linguistics	9
Country	10
Capital city	10
Food	10
Irrelevant general	20
Total	81

Table 1: Document collection summary

1.2 Question Dataset

Three question files were created for evaluation. Questions had four options and one correct answer. Each question focused on a single page.

Dataset	#	Description
`questions0`	20	General themed questions
`questions1`	25	Specific questions
`questions2`	25	Niche questions

Table 2: Document-based datasets used for evaluation.

All questions were generated using ChatGPT 5.5 [2] and saved in JSONL format. Each dataset was designed to be harder and more dependent on retrieval.

2 Method

2.1 Baseline Model

The baseline system runs `Gemma2:2b` offline using Ollama without retrieval. Only the question and answer choices were passed to the model.[3]

2.2 RAG System

The RAG system used the base system + BM25 retrieval.

Wikipedia pages were split into overlapped chunks. The top retrieved chunks were added to the end of prompt[4] before generation. Chunks from the same document were merged before adding.

2.3 Settings

Several retrieval settings were tested[5]. The best setting was selected based on accuracy. Different retrieval settings produced different results depending on the dataset.

For the more "difficult" `questions2` dataset, several configurations achieved perfect accuracy. Larger chunk sizes generally performed better on highly document-specific questions.

However, for `questions1`, performance differences between configurations were more visible. Smaller chunk sizes with larger retrieval depth often performed better than very large chunks.

[1] See Appendix A for a full list of page titles.

[2] See Appendix B for the prompts for question generation.
[3] See Appendix C for the basic prompt template.
[4] See Appendix C for the prompt template of RAG system.
[5] See Appendix D for accuracy scores

This suggests that retrieval performance depends strongly on:

- chunk size
- retrieval depth
- question specificity

3 Results

Dataset	Baseline (%)	RAG (%)	Δ
questions0	95	95	+0
questions1	72	92	+20
questions2	36	100	+64
Merged total	65.7	95.7	+30

Table 3: Accuracy across the three evaluation datasets.

The RAG system showed larger improvements on the harder datasets, where retrieval became more important. See Table 3.

4 Error Analysis

Two main error types were observed during testing: retrieval and reasoning failure. Error examples are included in Appendix E.

4.1 Retrieval Failure

Some questions failed because the correct document chunk was not retrieved.

For example, in a question comparing *güllaç* and baklava, the retrieved chunk only discussed baklava and did not mention güllaç.

Some retrieved chunks, however, were completely unrelated to the question. This happened mostly when the target document does not have explicit keyword match with the question or when document chunk sizes are small. There is also a case were the query has important keywords that may be out-of-vocabulary for the BM25 retriever[6].

4.2 Reasoning Failure

Some questions failed even when the correct information was retrieved.

For example, the model confused the direction of grammatical change from Old Persian to Middle Persian, incorrectly reversing the shift from synthetic to analytic grammar.

[6]This hypothesis is further supported by the fact that two examples in `question1` dataset were consistently failing in retrieval stage: the question about gullac and the question about the Gorno-Badakhshan region.

4.3 Wrong output token

At first the RAG model would output words or sentences instead of letters. Changing the prompt fixed every instance of this error.

4.4 Improvements

Several small improvements were tested:

- increasing chunk overlap
- filtering out very small chunks
- increasing top-k retrieval
- balancing top-k and chunk size
- improving the prompt

These changes improved the system performance to almost perfect accuracy.

5 Conclusion

This assignment showed that retrieval improves question answering performance noticeably. The RAG system achieved higher accuracy than the baseline model. Despite only using a 2B-parameter LLM, the retrieved documents did not distract the model.

The experiments also showed that retrieval quality is very important. Missing or unrelated chunks often caused the wrong answers.

Limitations

The dataset was small and based only on Wikipedia pages. BM25 retrieval also depends heavily on keyword overlap and may fail on semantic similarities. The model also was only two billion parameters, with limited knowledge and capabilities.

Acknowledgments

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A Wikipedia Pages

The full document collection is listed below.

A.1 Language

- Avestan
- Balochi language
- Gilaki language
- Iranian languages#Old Iranian
- Kurdish language
- Luri language
- Mazanderani language
- Middle Persian
- Ossetian language
- Pashto
- Persian language
- Tat language (Caucasus)
- Talysh language
- Zaza language

A.2 Language Family

- Caspian languages
- Category:Northwestern Iranian languages
- Eastern Iranian languages
- Indo-European languages
- Indo-Iranian languages
- Iranian languages
- Proto-Indo-European language
- Western Iranian languages

A.3 Linguistics

- Dialect continuum
- Language contact
- Language family
- Languages of Iran
- Languages of the Caucasus
- Morphology (linguistics)
- Persian alphabet
- Phonology
- Syntax

A.4 Country

- Afghanistan
- Armenia
- Azerbaijan
- Georgia (country)
- Iran
- Iraq
- Pakistan
- Russia
- Tajikistan
- Turkey

A.5 Capital City

- Ankara
- Baghdad
- Baku
- Dushanbe
- Islamabad
- Kabul
- Moscow
- Tehran
- Tbilisi
- Yerevan

A.6 Food

- Aush reshteh
- Baklava
- Borscht
- Chapli kebab
- Chelow kabab
- Dolma
- Ghormeh sabzi
- Kabuli palaw
- Khachapuri
- Pilaf#Central_Asia

A.7 Irrelevant General

- Architecture
- Basketball
- Bicycle
- Chess
- Chocolate
- Cinema
- Coffee
- Computer mouse
- Honey
- Mount Everest
- Ocean
- Origami
- Photosynthesis
- Piano
- Quantum mechanics
- Rainforest
- Shark
- Solar System
- Volcano
- World War II

B Question Generation Prompts

B.1 Prompt for questions0.jsonl

```
Find twenty Wikipedia pages related to
ten additional languages.

Also include twenty pages unrelated to
linguistics, along with pages
about the countries, capitals, and food
associated with those languages.

Organize the page titles into a
dictionary grouped by topic
(language, linguistics, food, capital
city, country, etc.).

Then generate twenty four-choice
multiple-choice questions based on
these documents for evaluating a RAG
system. Include the correct answer
for each question.

Export the dataset in JSONL format.
```

B.2 Prompt for questions1.jsonl

```
the questions you generated were very
simple. the questions should be
about very specific segments and niche
information inside the documents.
i am attaching some documents to start
with. Generate 5 questions for
each one. questions should be only
answerable by having access to that
specific document:
```

B.3 Prompt for questions2.jsonl

```
make even more niche questions, that a
normal pretrained gemma 3 doesn't
know about.

here are more documents. export in jsonl
```

C Prompts

C.1 Baseline Prompt

```
Respond to the question with a single
letter: A,B,C or D.
Only output the letter of the correct
answer.

Question: {question}

Answer:
```

C.2 RAG Prompt

```
Answer the multiple-choice question
using the retrieved documents as the
main evidence.

Retrieved documents:
{documents}

Question:
{question}

Output only one letter: A, B, C, or D.
Exactly one character from this set: A B
C D.
Do not write words. Do not explain.

Answer:
```

D Retrieval settings

This appendix lists the retrieval configurations tested during the RAG experiments. The columns q0, q1, and q2 refer to `questions0`, `questions1`, and `questions2`. Missing values indicate that the setting was not tested on that dataset. See Table 4.

Top-k	Chunk size	Overlap	q0	q1.	q2
5	250	50	90.0	92.0	96.0
3	450	80	85.0	92.0	100.0
2	250	50	–	84.0	92.0
2	600	90	–	88.0	100.0
1	900	100	–	84.0	100.0
4	600	100	95.0	–	–

Table 4: Retrieval settings tested across the three question datasets

E Example Error Cases

This appendix presents representative failure cases from the evaluation.

E.1 Retrieval Failure Example

```
QUESTION:
What distinguishes the dessert gullac
from baklava in preparation?

A: Uses honey instead of syrup
B: Uses milk-soaked layers instead of
baked pastry layers
C: Contains no nuts
D: Is fried instead of baked

DOCUMENTS

# baklava

On the island of Lesbos in Greece a type
of baklava is still known as placenta
(Greek: platsenta), which is the name of
an Ancient Greek pastry that is often
seen as the predecessor of baklava. The
latter is a baked dessert with very
thinly made pastry layers and chopped
nuts. The base for this modern placenta
is made with leaves of filo dough, and
nuts stacked upon each other. After
baking, it is soaked in simple syrup and
sprinkled with cinnamon.

answered: A
correct: B
```

The retrieved chunk discussed only baklava and did not contain information about güllaç. The chunk was from a related document topic, and it made a comparison between baklava and another pastry. These similarities confused both the retriever and the LLM.

E.2 Reasoning Failure Example

```
QUESTION:
What major grammatical shift occurred
from Old Persian to Middle Persian?

A: Analytic to synthetic
B: Synthetic to analytic
C: Agglutinative to fusional
D: Isolating to agglutinative

DOCUMENTS

# middle persian

The elision of unstressed word-final
syllables during the transition from Old
to Middle Persian has eliminated many
grammatical endings.
As a result, compared to the synthetic
grammar of Old Persian, Middle Persian
belongs to a much more analytic language
type.

answered: A
correct: B
```

The correct information was retrieved, but the model reversed the direction of grammatical change.

E.3 Missing Keyword Retrieval Example

```
QUESTION:
Which writing system feature uniquely
characterizes Pahlavi scripts?

A: Use of vowels only
B: Strict phonetic representation
C: Use of Aramaeograms
D: Absence of ligatures

DOCUMENTS

# middle persian

In spite of the availability of signs
for each sound, Manichaean spelling did
not always make perfectly phonetic use
of them.

answered: B
correct: C
```

The retrieved chunk discussed phonetics and spelling but did not contain the key concept "Aramaeograms".

E.4 Unrelated Retrieval Example

```
QUESTION:
Which linguistic diversity is noted in
the Gorno-Badakhshan region?

A: Turkic-only languages
B: Slavic dialects
C: Multiple Pamir languages alongside
Tajik
D: Only Persian dialects
```

```
DOCUMENTS

# iranian languages old iranian

The multitude of Middle Iranian
languages and peoples indicate that
great linguistic diversity must have
existed among the ancient
speakers of Iranian languages.

answered: D
correct: C
```

The retrieved document was only loosely related to the topic and did not contain the information required to answer the question. It was most likely retrieved due to the existence of *"linguistic diversity"* keyword.

Tokenizer blindess Depending on the retriever settings (chunk size and overlap, top-k), different documents will be retrieved but none will match the target document or even the word *Gorno-Badakhshan* in the first place. It is likely that the query of BM25 algorithm skips on this out-of-vocabulary word completely.